Consequences of mean motion resonances in AGN disks on the merging black hole properties

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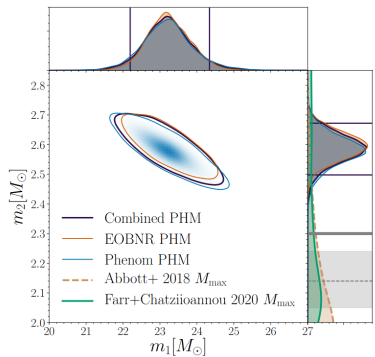
University of Western Australia / OzGrav

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Moncrieff et al. (2026), MNRAS 545, staf2217 • Moncrieff & Panther (2025), MNRAS 543, 1833 • Moncrieff et al. (in prep.)

Gravitational-wave astronomy

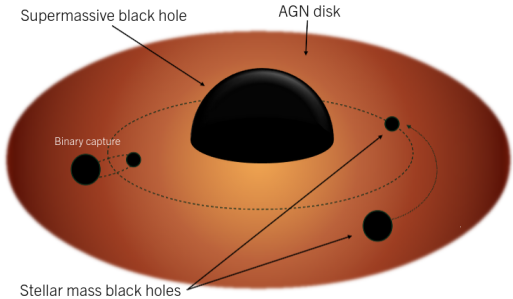
- LIGO–Virgo–KAGRA: **hundreds of merging stellar-mass black hole binaries** detected.
- GW strain gives BBH masses, spins, mass ratio, etc.
- Very poor sky localisation, so no host galaxy association $\Rightarrow$ formation channel must be inferred from the *population*.
- E.g. **GW190814**, $q = m_2/m_1 \approx 1/10$: the most extreme mass ratio event detected - hard to form via standard formation channels.



GW190814 extreme mass ratio BBH merger, from Abott et al. 2019

Hierarchical mergers in AGN disks

- Proposal to explain "unusual" events, difficult to produce via traditional pathways.
- BHs in an AGN disk migrate under gas torques and accumulate at migration traps.
- There they pair into binaries and merge rapidly.
- Repeated (hierarchical) mergers build mass and can skew the mass ratio.

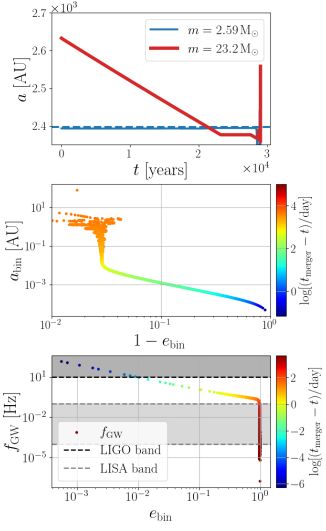


Cartoon of binary capture in an AGN disk.

Check with N-body simulations

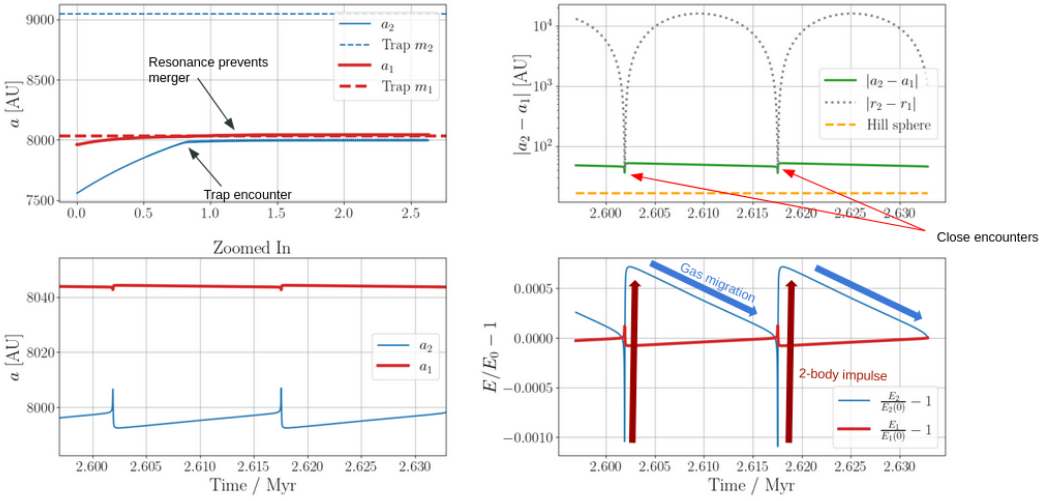
- Studies of BBH merger populations in AGN are generally very simplified.
- Test assumptions with the relativistic N-body code **TSUNAMI** coupled to a (Sirko Goodman) AGN disk with migration torque + gap opening prescriptions.
- *Certain* AGN disks indeed produce high mass ratio mergers like GW190814 at traps.

But: several *often overlooked* obstacles prevent this in general...



Moncrieff et al. 2026

Barrier to binary capture: resonance blocking

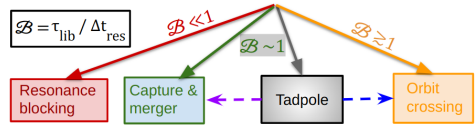


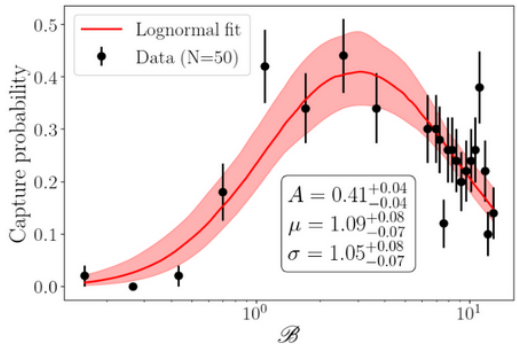
Moncrieff et al. (2026)

One number decides the outcome: the adiabaticity parameter $\mathscr{B}$

$$\mathscr{B} \equiv \frac{\tau_{\rm lib}}{\Delta t_{\rm res}} = \frac{\text{resonance libration time (at } \Delta a = 2\sqrt{3}R_{\rm Hill})}{\text{resonance crossing time (differential migration)}}$$

- The **adiabaticity of resonance crossing**.
- $\mathscr{B} \ll 1$: adiabatic $\rightarrow$ resonance capture $\rightarrow$ blocking.
- $\mathscr{B} \gg 1$: resonance jumped $\rightarrow$ close encounter unlikely (fast pass by).
- $\mathscr{B} \sim 1$: No resonance capture, close encounter $\rightarrow$ likely binary capture.
- **Analytic** in $(m_1, m_2$, disc model, radius), matches sims well.

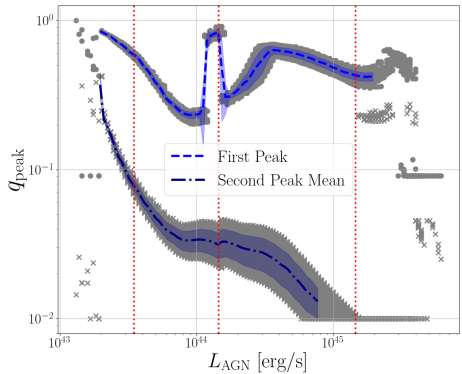




Which AGN make GW190814s? MMRs as a mass-ratio filter

- MMRs **select** which mass-ratio encounters can end in capture/merger.
- Preferred merger q (at trap locations) **correlates strongly with AGN luminosity**.
- Narrow luminosity band peaks at $q \approx 0.1$: **GW190814-like**.

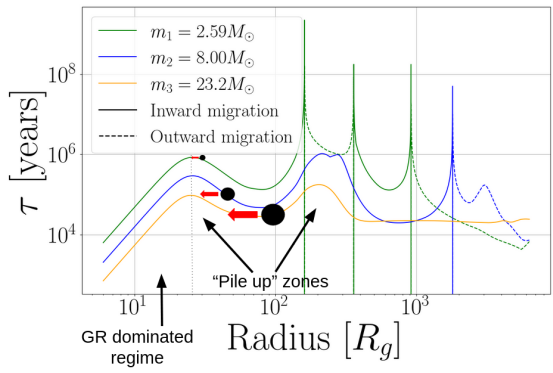
$\Rightarrow$ possible route to **ranking AGN host galaxies** by expected merger demographics **without an EM counterpart**.



Moncrieff et al. (2026)

The fate of the survivors: resonant chains migrate inward

- Gap opening *can* result in resonance chain migrating past the trap, toward the SMBH.
- Destination: the region where **general relativity dominates the dynamics**.
- Possible channel for **Extreme Mass Ratio Inspirals**.

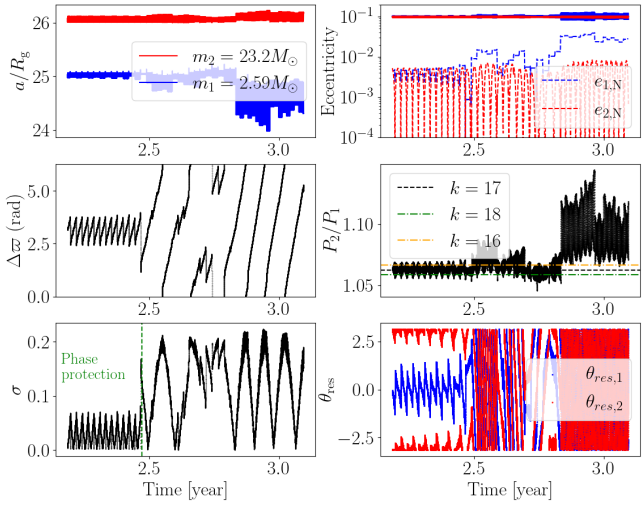


Moncrieff et al. (in prep.)

Relativistic mean-motion resonances

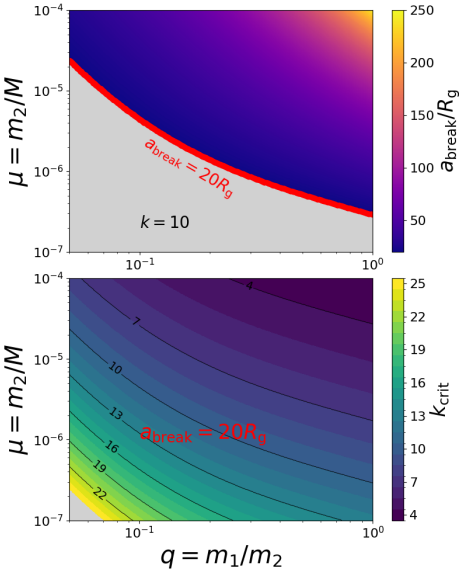
General outcome:

- 1PN apsidal precession increases **effective eccentricity** $\sigma = ||\vec{e}_1 + \vec{e}_2||$
- And hence resonance width $\Longrightarrow$ **MMR overlap**.
- Onset of chaos $\rightarrow$ strong scatter or even binary capture.



Which chains survive to the EMRI regime?

- For a $k : k-1$ resonance, break radius a_{break} depend on $q = m_1/m_2$ and $\mu = m_2/M$.
- Higher k resonances break first; near the SMBH only low k resonances survive to EMRI regime.
- Chain EMRIs could be observable with upcoming LISA detector.



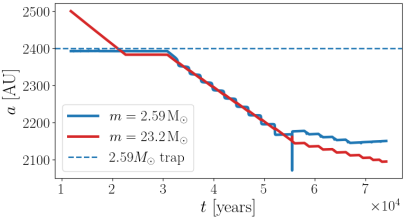
Summary

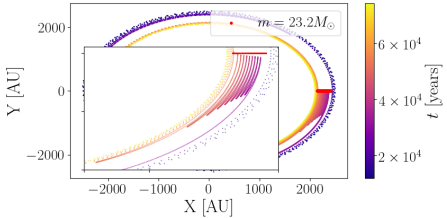
1. BHs in AGN discs migrate like planets; **MMRs control encounter outcomes**.
2. One analytic adiabaticity parameter $\mathscr{B}$ predicts blocking vs merger vs pass by, giving insight into population level impact of MMRs and disk properties on BBH mergers.
3. Preferred merger q **correlates with AGN luminosity** (Moncrieff et al. 2026, *Not all roads lead to merger: AGN disc properties influence the interactions of highly unequal mass black holes*)
4. Surviving resonant chains can migrate inward until **GR induced chaotic MMR overlap breaks the resonance** $\Rightarrow$ new pathway to **EMRIs and relativistic mergers** (Moncrieff et al. in prep.).

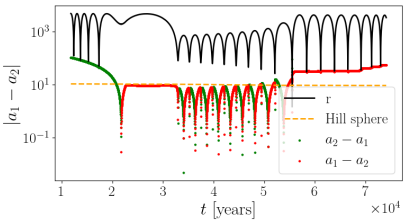
Thank you!

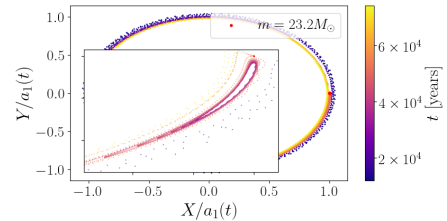
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Backup: tadpole / co-orbital resonances

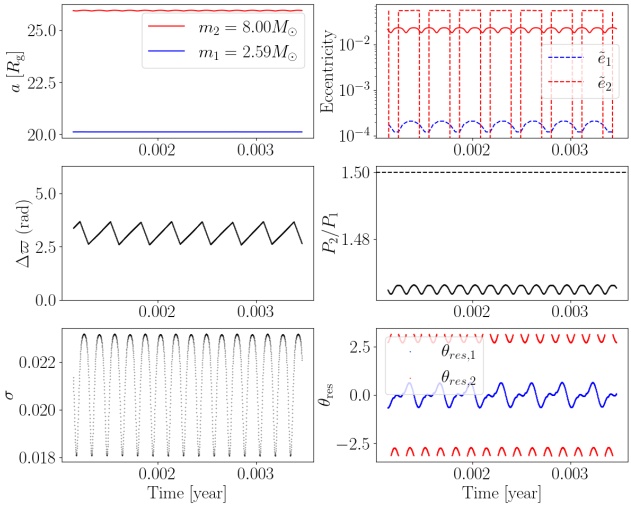








Backup: Stable MMR in EMRI regime



Backup: Strong scatter and capture

